

Homework 7

Due: Wednesday 5/7 at 11:59pm

Complete the following problems and submit them through Google Classroom. You can submit the picture(s) of a written file, a Google Doc, or a PDF. (Note that you can attach multiple files when making a submission.) Name your file(s) based on your name or Knox login.

1. (9 points) Consider the following maze in the format we used for Lab 5:

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WWWWWWWWW
WT.WW...TW
WW.W..WW.W
W..S..T..W
W.WWWWW.W
W...W....W
W.....WWW
WWWWWWWWW
```

This maze has 3 treasures. We name them top left, top right, and right based on their positions. Assume that we use DFS as implemented in the lab. Give a search order (ordering of the directions up (U), down (D), left (L), and right (R)) that visits these treasures in each of the following orders:

- (a) right, top right, top left
 - (b) top right, right, top left
 - (c) top left, top right, right
2. (8 points) Using the implementation of binary search tree operations we discussed in class, draw the trees that result from the following operations:
 - (a) Inserting 142, 175, 25, 38, 205, 9, 12, 155, 300, 50, 79, and 127 into an initially-empty tree (in that order). In order to ensure that you get partial credit, I suggest that you draw the tree after each of these insertions.
 - (b) Deleting 142 from the tree you drew for part (a).
 3. (7 points) Draw the unique binary tree that has an preorder traversal of 1, 2, 3, 4, 5, 6, 7, 8 and an inorder traversal of 2, 3, 1, 5, 4, 7, 8, 6. Note that this will not necessarily be a binary search tree.
 4. (1 point) Add a statement about collaboration to your submission for the first problem.